UNIT 1 · KINEMATICS · WEEK 4

Ball Drop — Measuring *g*

Physics 2026–27 · in class Tuesday, October 27 · record your drop over the weekend

NAME _____ DATE _____

Why this one is different. In the cart lab you measured an acceleration nobody knew in advance. Here the answer is known: $g = 9.8\ m/s^2$ downward. Measuring a quantity whose true value you already know is how you find out what your *method* is worth — camera, frame rate, tape, reaction time. This is **in-class practice (CW)**, not a second formal report: the analysis is the same one you used for Lab Report I, just aimed at a known target.

Objectives

WHAT YOU SHOULD BE ABLE TO DO BY THE END

- Capture an object in free fall with a phone camera and read its position frame-by-frame.
- Turn frame numbers into times using the video's frame rate.
- Plot position vs time and position vs t^2, and identify which one is linear.
- Extract g from the slope and compare it to $9.8\ m/s^2$ with an honest uncertainty.

What you need

PHONE + TAPE

- A phone or camera that can record video and play it **frame-by-frame** (QuickTime on a Mac, VLC on any platform, or your phone's editor).
- A measuring tape, or tape marks every **0.10 m** — a measuring tape visible in the shot is easiest.
- An object to drop: a marble, a small ball, a battery, a grape — anything small and heavy enough to fall cleanly.

Setup & sign convention

DECIDE ONCE, THEN STAY CONSISTENT

Up is positive. The origin (`d = 0`) is at the floor. You will drop the object from `d = 1.0 m` straight down to the floor. So the position `d` starts at `1.0 m` and *decreases*, and the acceleration is *negative*: `a = −g`.

1. Tape a **1.0 m** drop height on a wall, a doorway, or a tape measure — one clear height, marked at the release point.
2. Put a ruler or tape in the frame with **0.10 m marks**, so you can read the object's position from the video.
3. Set the camera to its **highest frame rate** (960 fps slow-mo is ideal, 240 fps is great, 60 fps works). Record from the side, and do not move the camera.

ASK YOURSELF

Does it matter whether we call down positive? Does it matter where we put the origin? You should be able to answer both — and show that the *value* of `g` does not change either way.

Procedure

FRAMES TO TIMES TO POSITIONS

1. **Find the frame rate, `f` (fps).** Check your phone's camera settings/app, or count frames between two events a known time apart. The time between frames is `τ = 1/f` seconds.
2. **Open the video frame-by-frame.** Call the release frame `Δframe # = 0`.
3. **Step forward and record the frame number** the instant the object passes each 0.10 m mark, from `d = 1.0 m` down to `d = 0 m`.
4. **Fill the table.** Time is `t = (Δframe #) × τ`. Then add t^2 and the measured `d`.
5. **Repeat the drop 2–3 times** if you have time, and average. One drop is a measurement; three drops tell you how repeatable it is.

BLURRY FRAMES ARE NORMAL — READ THE CENTER OF THE BLUR. REPORT ALL VALUES IN SI UNITS (M, S).

Data

RECORD WHAT YOU ACTUALLY SEE

FRAME RATE F (FPS)	TIME BETWEEN FRAMES T = 1/F (S)

ΔFRAME #	T = (ΔFRAME #) × T (S)	T^2 (S^2)	D (M)
0	0	0	1.00

Graphs

ONE CURVED, ONE STRAIGHT

Make both graphs (by hand on graph paper, or in Desmos/Sheets):

i. Position d as a function of time, t.

ii. Position d as a function of time squared, t^2.

The equation governing the fall, with up positive and $v_0 = 0$, is $d = d_0 + \frac{1}{2} a t^2$. Only one of your two graphs is a straight line.

Analysis

SLOPE → ACCELERATION

2.3 Best-fit line

LINEAR GRAPH ONLY

Draw a best-fit line on the **linear** graph — a single straight line that passes *near* the points, not connect-the-dots. Submit that graph with the line included.

2.4 Extract g

UNITS THE WHOLE WAY

a. Compute the **slope** of your best-fit line, with units. (Hint: compare $d = d_0 + \frac{1}{2}at^2$ to $y = b + mx$.)

b. Use the slope to compute the object's **acceleration**, a. Watch the sign — up is positive, so if it fell, what sign should a have?

c. The accepted value is $g = 9.8\ m/s^2$ downward. Does your value agree? By how much is it off, in percent?

d. Name **two** sources of uncertainty (frame-rate error, reading a blurry frame, parallax, air resistance, the tape not being vertical…) and say which way each might push your result.

SLOPE & WORK

2.5 Explain the shape

CONCEPT

In one or two sentences: why is d vs t curved, while d vs t^2 is straight? Use the word **proportional**.

YOUR ANSWER

Graded as formative classwork (CW), on the homework 4–3–2–1 scale: complete the table, both graphs, the slope-to-g calculation, and the uncertainty note. The full seven-section formal report is the cart lab — this one is the rehearsal with a known answer.